

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE(AI)	Date:	

Code of Conduct

/ T. Kiran Kumar (192472321) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.Kiran Kumar

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether **"to the car"** is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates **"with AC sleeper"** with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations

- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

1. AI-POWERED INSURANCE CLAIM PROCESSING SYSTEM

a) Analyze the structural ambiguity using CFG

The sentence is:

“I reported the damage to the car after the accident.”

The phrase **“to the car”** can have different attachments.

Interpretation 1:

“I reported [the damage to the car] [after the accident].”

Here, **“to the car”** describes the damage.

Interpretation 2:

“I reported [the damage] [to the car] [after the accident].”

Here, **“to the car”** is attached to the reporting action.

A CFG can generate both structures because prepositional phrases can be attached to different parts of the sentence.

b) Limitations of basic CFG

A basic CFG has several limitations:

- It can generate multiple ambiguous parses.
- It does not naturally represent subject–verb agreement.
- It may become inefficient for long sentences.
- It does not assign probabilities to competing interpretations.
- It has difficulty handling incomplete conversational input.

c) Improved parsing approach

A better system can combine:

- **PCFG:** Assigns probabilities to different parses and selects the most likely interpretation.
- **Feature Structures:** Enforce agreement such as number, person, and tense.
- **Earley Parsing:** Efficiently handles ambiguity and incomplete input.

The architecture can be:

Customer Input → Tokenization → Earley Parser → Feature Checking → PCFG Ranking → Semantic Role Extraction → Claim Information

d) Benefits

The proposed architecture improves:

- **Accuracy:** PCFG selects the most probable interpretation.
- **Reliability:** Feature structures reduce grammatical errors.
- **Scalability:** Earley parsing can handle complex and partial sentences.
- **Automation:** Important claim details can be extracted more accurately.

Thus, the system can reliably identify the claimant, damaged object, cause, and claim amount.

2. INTELLIGENT RAILWAY TICKET BOOKING ASSISTANT

a) Analyze the possible syntactic interpretations using CFG

The sentence is:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

The main structure is:

Book → two tickets → from Chennai → to Delhi → for my parents → with AC sleeper

The phrase **“with AC sleeper”** can potentially be attached to the passengers or the tickets.

The intended interpretation is:

Book [two tickets] [from Chennai] [to Delhi] [for my parents] [with AC sleeper].

Here, **AC sleeper** describes the ticket/class, not the passengers.

b) Problems with top-down parsing

Top-down parsing starts from the start symbol and tries possible grammar rules. When several structures are possible, it may repeatedly backtrack.

This can cause:

- Excessive backtracking
- Slow processing
- Difficulty with ambiguous phrases
- Problems with incomplete requests
- Increased response time for long queries

These problems are important in conversational railway booking systems where users expect quick responses.

c) Earley Parsing

Earley parsing uses a chart to store partial parsing results. It can handle:

- Ambiguous structures
- Incomplete input
- Long-distance dependencies
- Multiple possible parses

For example, while the user is typing **“Book two tickets from Chennai to...”**, Earley parsing can maintain a partial parse instead of waiting for the complete sentence.

d) Comparison

Feature	Top-Down Parsing	CFG Parsing	Earley Parsing
Ambiguity	Limited	Can generate alternatives	Handles alternatives efficiently
Backtracking	High	Possible	Reduced through chart parsing
Partial input	Poor	Limited	Good
Complex queries	Less efficient	Moderate	Better
Real-time use	Limited	Moderate	Suitable

Therefore, **Earley Parsing** is more suitable for a conversational railway booking system because it can handle incomplete and ambiguous passenger requests efficiently.

3. AI-BASED LEGAL CONTRACT ANALYSIS SYSTEM

a) Analyze the syntactic structure

The sentence is:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

The main structure is:

The supplier → must replace → the defective components → within thirty days

The relative clause is:

“who delivered the equipment to the company”

Its structure is:

who → delivered → the equipment → to the company

The phrase **“to the company”** is associated with **delivered**, meaning that the supplier delivered the equipment to the company.

b) Limitations of basic CFG

Basic CFG can represent the grammatical structure, but it has difficulty with:

- Nested clauses
- Relative clauses
- Long sentences
- Multiple prepositional phrases
- Ambiguous attachments
- Semantic relationships

It may therefore incorrectly associate **company** with the delivery action or assign responsibility to the wrong entity.

c) Proposed architecture

The system can use:

Legal Text → Tokenization → CFG Parsing → Earley Parsing → Feature Structures → PCFG Ranking → Semantic Role Analysis → Information Extraction

- **CFG:** Represents basic grammatical structures.
- **Earley:** Handles complex and incomplete structures.
- **Feature Structures:** Enforce grammatical agreement.
- **PCFG:** Selects the most probable interpretation.
- **Semantic analysis:** Identifies legal roles and obligations.

d) Step-by-step information extraction

1. **Supplier:** Identified as the subject and responsible party.

2. **Action:** “must replace”
3. **Equipment:** Object mentioned in the relative clause.
4. **Company:** Recipient of the equipment.
5. **Obligation:** Supplier must replace defective components.
6. **Deadline:** Within thirty days.

The final interpretation is:

Supplier → must replace → defective components → within 30 days

The proposed system reduces incorrect role assignment and improves automated contract analysis.

4. AI-BASED E-COMMERCE PRODUCT SEARCH SYSTEM

a) Analyze the syntactic ambiguity

The query is:

“Find laptops with a touchscreen for students under ₹60,000.”

The intended structure is:

Find [laptops] [with a touchscreen] [for students] [under ₹60,000].

Here:

- **laptops** → Product
- **touchscreen** → Product attribute
- **students** → Target user
- **₹60,000** → Price constraint

An incorrect parser might attach “**for students**” directly to “**touchscreen**”, changing the intended meaning.

b) Limitations of CFG

Basic CFG has difficulty with:

- Ambiguous phrase attachment
- Informal queries
- Missing words
- Elliptical expressions
- Long product descriptions
- Real-time processing

For example, users may enter incomplete queries such as “**laptops under 60k with...**”, which a basic CFG may not handle effectively.

c) Improved parsing architecture

The architecture can be:

User Query → Tokenization → Earley Parsing → Feature Structures → PCFG Ranking → Attribute Extraction → Product Search

PCFG selects the most probable parse, while feature structures identify properties such as product type, price, and target user.

The system can extract:

Product: Laptop

Feature: Touchscreen

Target user: Students

Price: Under ₹60,000

d) Benefits

The proposed system improves:

- **Search precision:** Correctly identifies user requirements.
- **Response time:** Earley parsing supports efficient partial parsing.
- **Accuracy:** PCFG helps select the most likely interpretation.
- **Customer experience:** Natural-language queries can be understood more effectively.

Therefore, the combined approach is more suitable for large-scale conversational e-commerce search.

5. INTELLIGENT AIRLINE VOICE ASSISTANT

a) Analyze the possible syntactic structures

The command is:

“Change my flight to Mumbai tomorrow with two passengers.”

The intended interpretation is:

Change [my flight] [to Mumbai] [tomorrow] [with two passengers].

Here:

- **Flight** → Booking
- **Mumbai** → Destination
- **Tomorrow** → Travel date
- **Two passengers** → Passenger count

An incorrect parser may attach **“with two passengers”** to **Mumbai** rather than treating it as passenger information.

The second command is:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

This contains a correction. The system should replace **Mumbai** with **Bangalore**.

b) Limitations of top-down parsing

Top-down parsing can have problems with spoken commands because:

- Speech may be incomplete.
- Users may correct themselves.
- Multiple interpretations may exist.
- Recognition errors can create unexpected words.

- Backtracking increases processing time.

This can lead to delayed or incorrect booking actions.

c) Proposed parsing architecture

A suitable architecture is:

Speech Input → Speech Recognition → Tokenization → Earley Parsing → Feature Structures → PCFG Ranking → Correction Handling → Booking Request

- **Earley Parsing:** Handles partial and incomplete commands.
- **PCFG:** Selects the most probable interpretation.
- **Feature Structures:** Validate information such as passenger count, destination, and date.
- **Correction handling:** Replaces previously identified information when the user corrects themselves.

d) Processing workflow

Step 1: Convert speech into text.

Step 2: Detect important entities such as destination, date, and passenger count.

Step 3: Use Earley Parsing to create possible syntactic structures.

Step 4: Use PCFG to select the most probable structure.

Step 5: Use feature structures to validate the extracted information.

Step 6: Handle corrections such as:

Mumbai → Bangalore

Step 7: Generate a structured booking request:

Action: Change flight

Destination: Bangalore

Date: Tomorrow

Passengers: 2

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis Professionalism	Sub. Total	Total %
1	4	4	4	3	3	18	94
2	3	3	4	4	4	18	
3	4	4	4	3	3	18	
4	4	4	3	4	4	18	
5	3	4	4	4	4	19	

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